SC1015 Mini-Project

Cell2Cell Telecom Churn Dataset

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THE TELECOM INDUSTRY

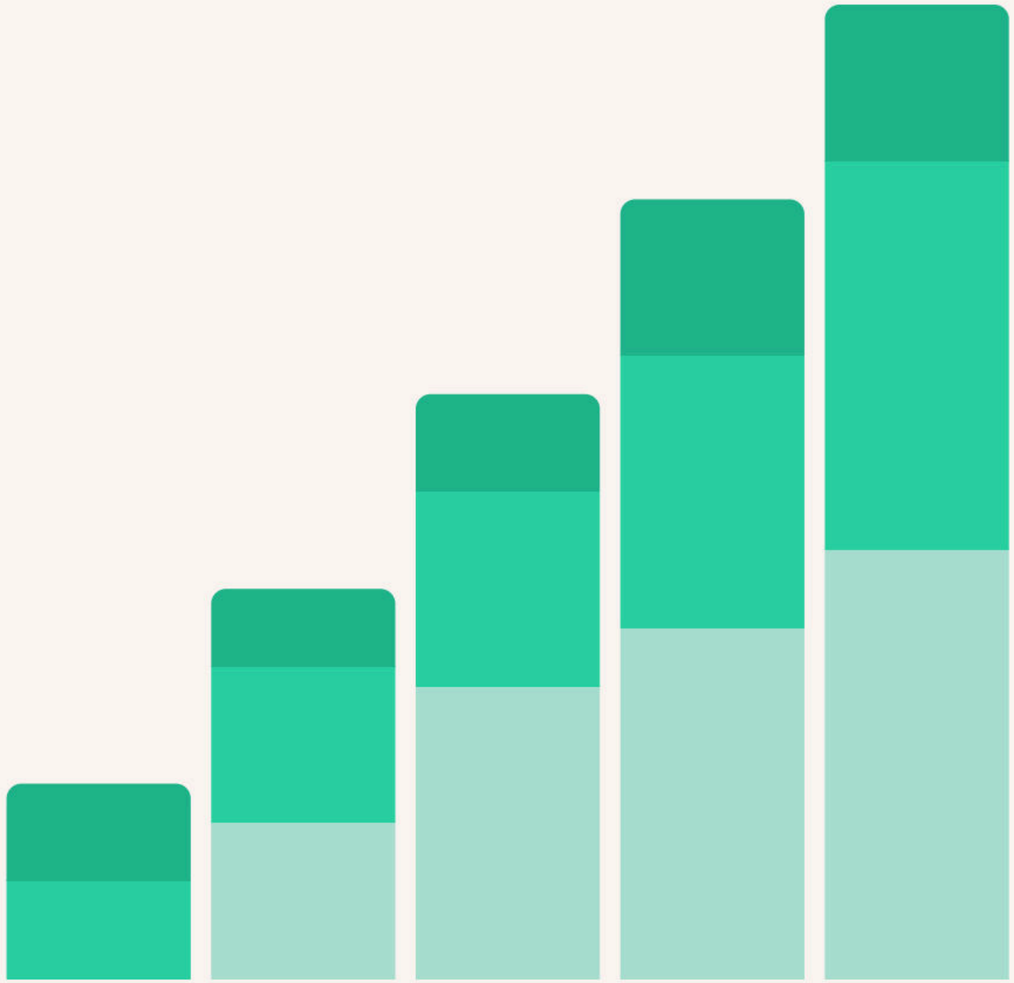
+6.2% annually

Revenue of USD1800 Billion in 2022 → USD2900 Billion by 2030

&

Over 8 Billion Mobile Subscribers

How can telecom companies survive in such a competitive market?



Problem

What are the variables affecting the churn rate of existing customers?

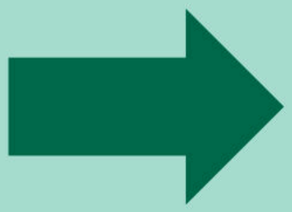
How effective are they in predicting churn rate?

WHAT IS CHURN?

-> Rate at which customers leave or discontinue their service with a telecom provider.

WHY CHURN?

High Churn Rates



Revenue Loss
Lower Competitive Advantage

Data Overview

CurrentEquipmentDays
RetentionCalls
TotalCallsInOut
PercChangeMinutes
MonthsInService'
OverageMinutes
CustomerCareCalls
MonthlyRevenue

TotalRecurringCharge
MonthlyMinutes
CreditRating
MadeCallToRetentionTeam
HandsetWebCapable
NewCellphoneUser
HasCreditCard
HandsetRefurbished

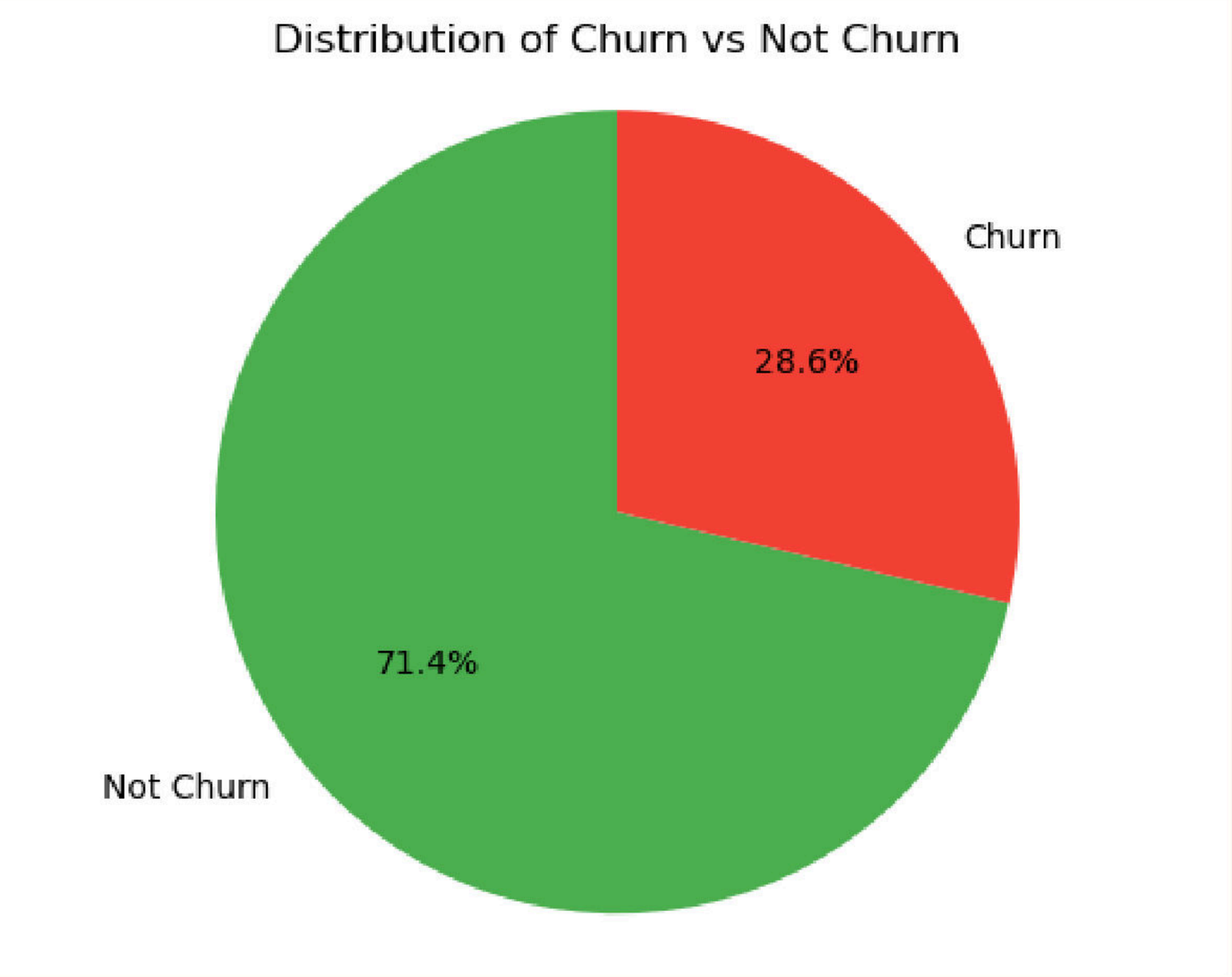
Data Cleaning

Remove Duplicates

Remove Null Values

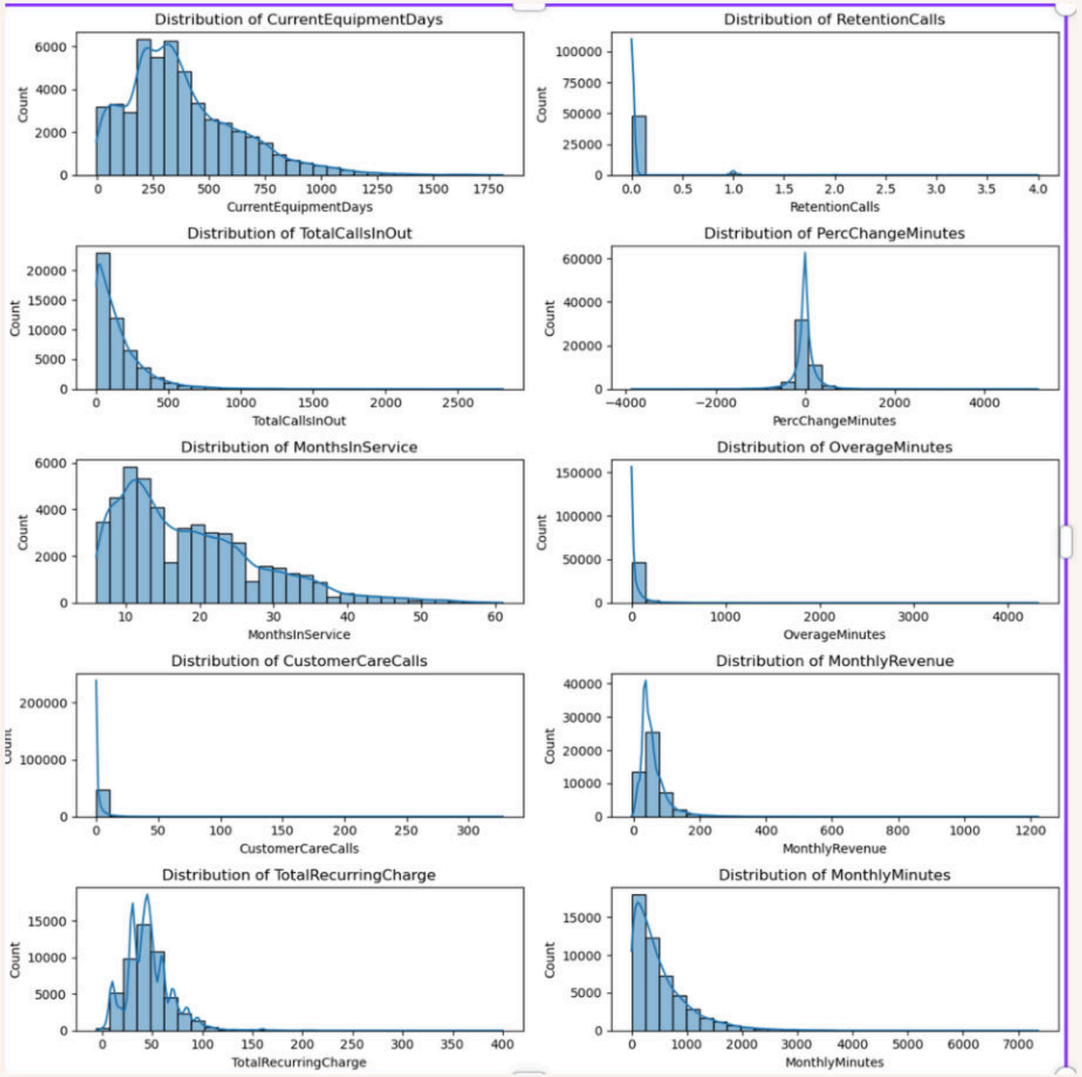
Binary Categorical to Numerical

Exploratory Data Analysis


Distribution of Churn vs Not Churn
Churn
28.6%
71.4%
Not Churn

Histogram Analysis

1. Most variables are right-skewed,
2. This indicate that for most of these features, a large segment of customers are clustered at the lower end of the spectrum with fewer customers having higher values. This can signal that the majority have similar, moderate usage or engagement, with a minority exhibiting higher usage or engagement.



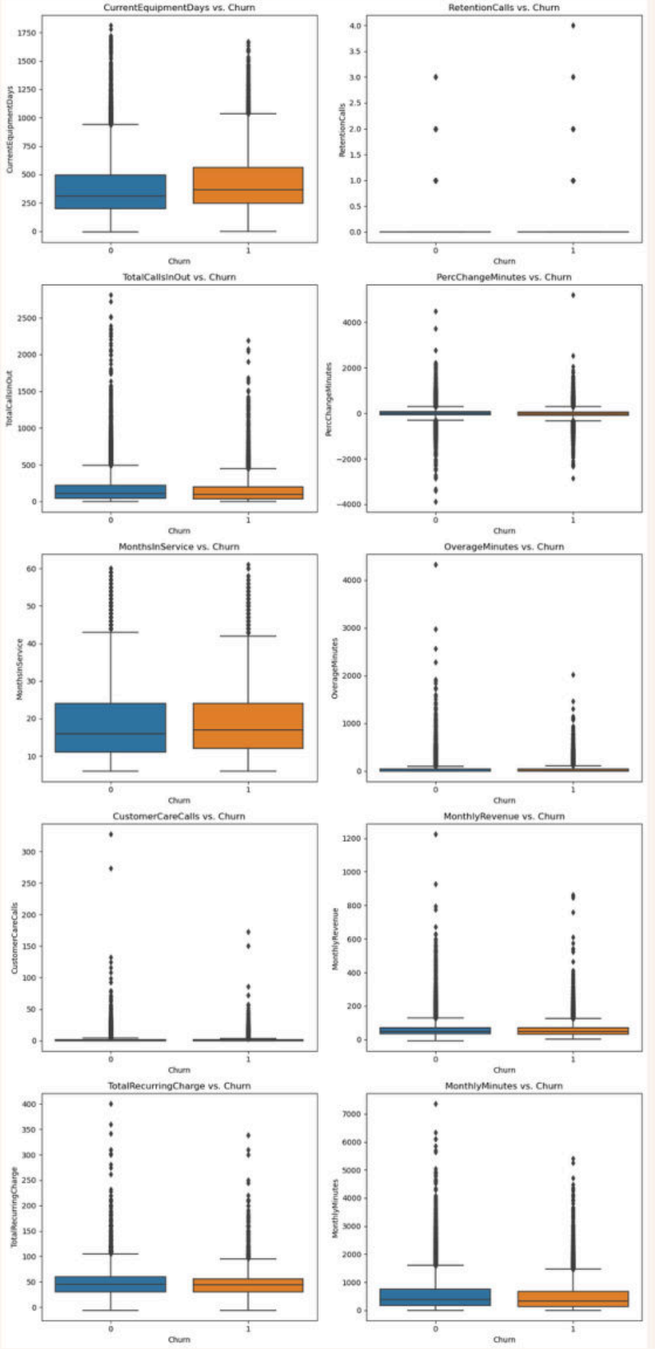
Box Plots

CurrentEquipmentDays: Customers who churned tend to have been using their current equipment for a shorter period than those who didn't churn, as seen by the median value of the boxplot for churned customers.

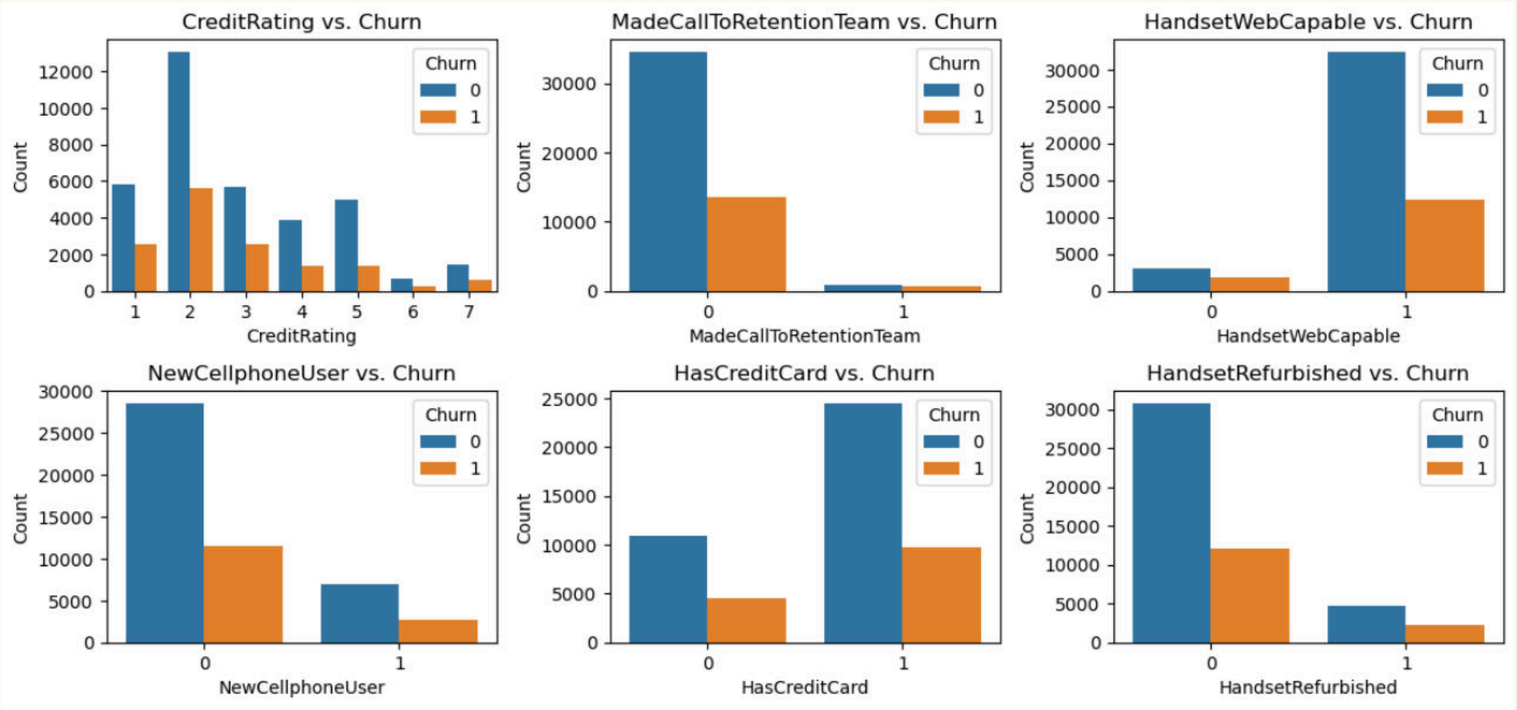
TotalRecurringCharge & MonthlyRevenue (finance-related): Churned customers often having lower charges and revenue

RetentionCalls: Churned customers have more retention calls

MonthlyMinutes & TotalCallsInOut (usage-related): Higher median values for churned customers



CatPlots



CreditRating: Varying churn rate across different credit ratings. Generally, lower credit rating seem to lead to higher churn rates

MadeCallToRetentionTeam: Higher proportion of customers who made a call to the retention team ended up churning

HandsetWebCapable: Churn seems to be lower among customers with web-capable handsets.

HandsetRefurbished: Customers with refurbished handsets show a higher churn rate.

HasCreditCard: Having a credit card does not show a stark difference in churn rates.

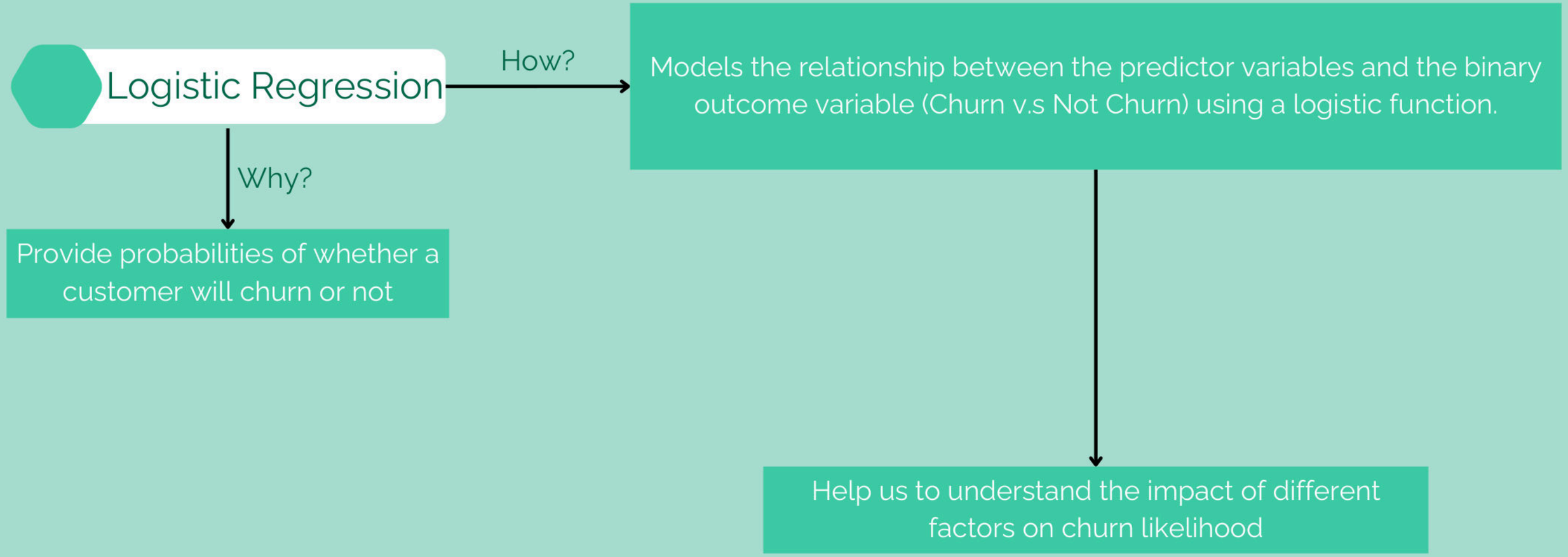
NewCellphoneUser: There's a small difference in churn between new cellphone users and more established users, with new users showing slightly more churn.

Machine Learning Utilisation

Feature Importance

- Which factors are significant in predicting churn?

Machine Learning Models



Machine Learning Models

Decision Tree

How?

Recursively split the feature space based on the most informative features, with each split maximizing the separation between churn and non-churn instances

Highlight decision making processes at each node

Identify the important features

Why?

Capture complex interactions between features and identify non-linear relationships that may influence churn.

Machine Learning Models

Random Forest

How?

Build multiple Decision Trees on random subsets of the training data and features

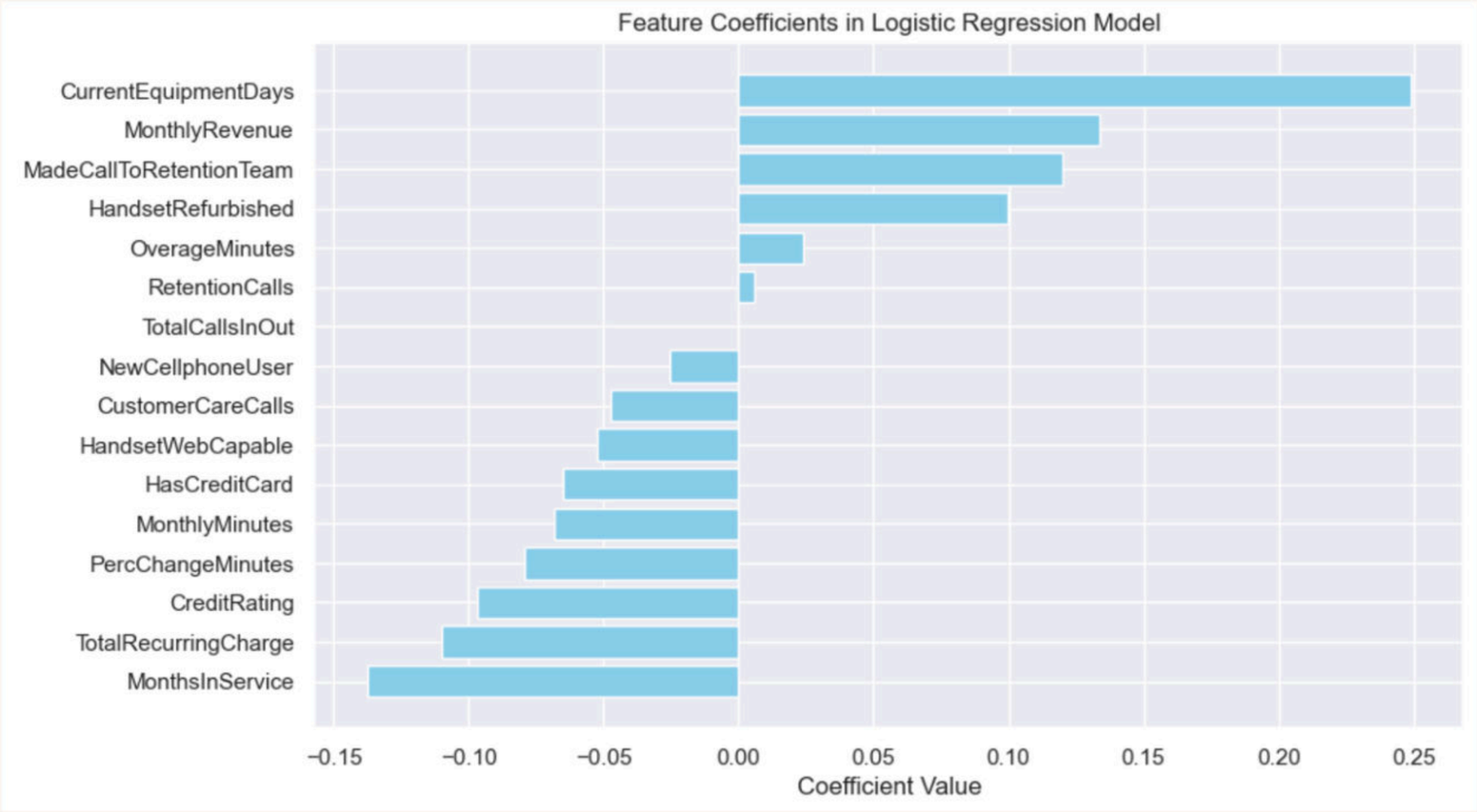
Why?

Offers improved robustness and predictive performance compared to individual Decision Trees

Leverage on the diversity of decision trees

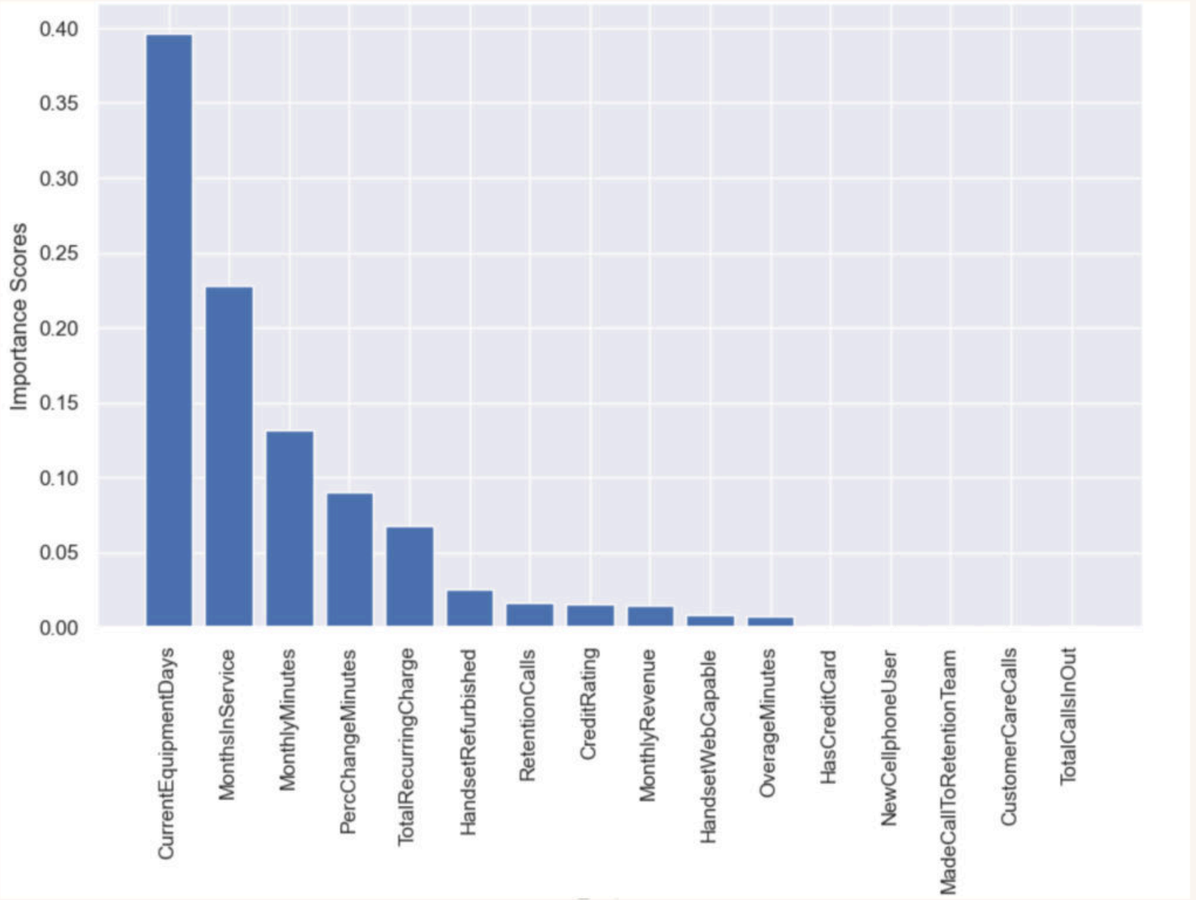
Provides insights into the relative importance of different features in predicting Churn

Logistic Regression



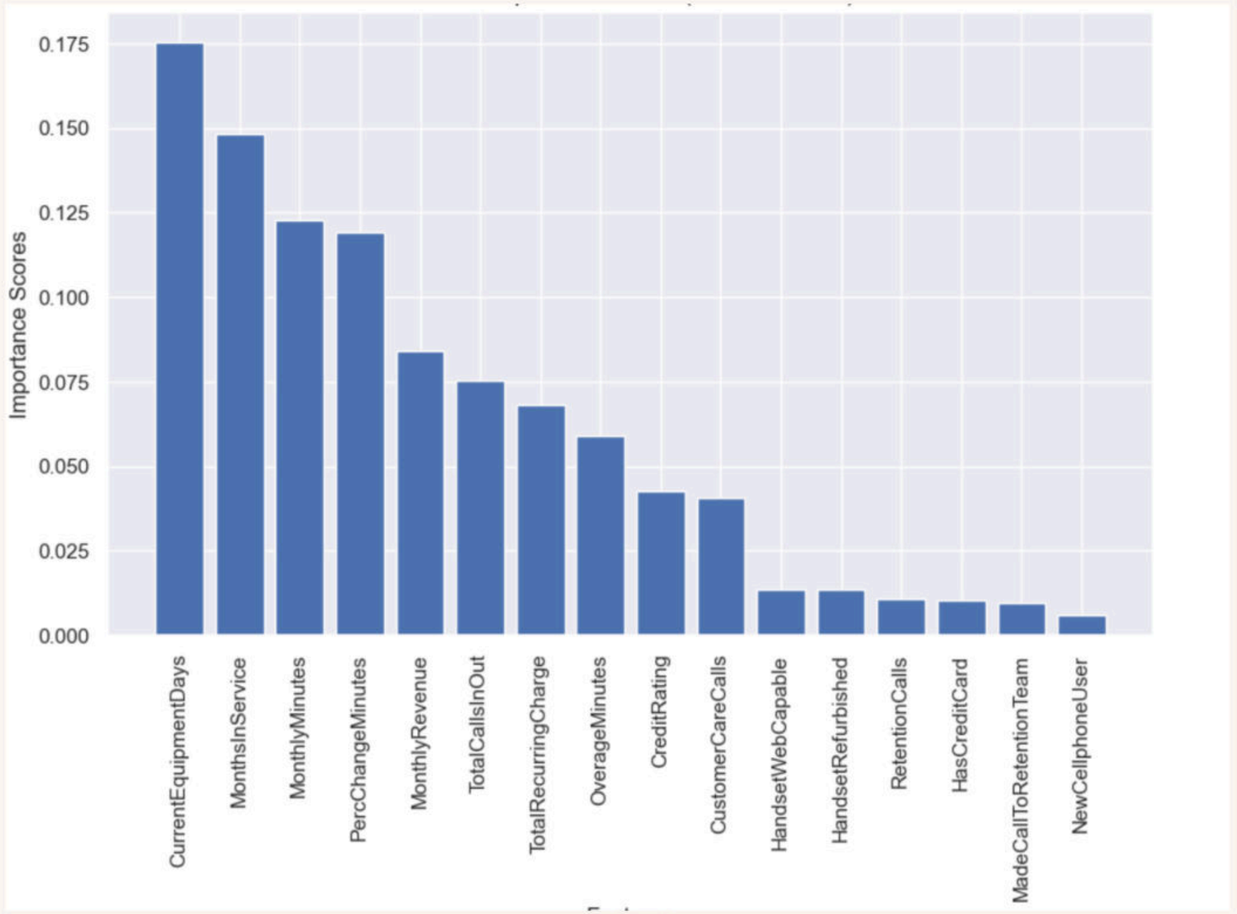
- The coefficient represents the impact of each feature on the likelihood of churn occurring.
- **CurrentEquipmentDays (0.25)** is the most influential feature which affect churn followed by **MonthsInService (-0.14)** and lastly, **MonthlyRevenue (0.13), MadeCallsToRetentionTeam (0.12) and TotalRecurringCharge(-0.110)**

Decision Tree



- The coefficient represents with how frequently a feature is used to make decisions at the internal nodes of the tree and how much it reduces impurities.
- **CurrentEquipmentDays (0.396)** was the most influential feature which affect churn followed by **MonthsInService (0.228)** thereafter **MonthlyMinutes (0.131).**

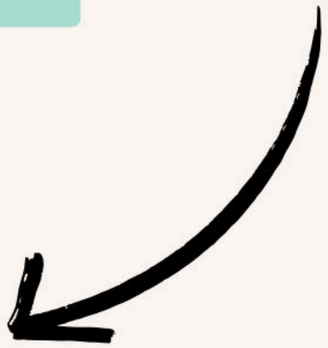
Random Forest



- The coefficient represents the relative importance of each feature in reducing impurity and making accurate predictions across the ensemble of trees.
- **CurrentEquipmentDays (0.175)** is the most influential feature which affect churn followed by **MonthsInService (0.148)** thereafter **MonthlyMinutes (0.123).**

Results

LOGISTIC REGRESSION + **DECISION TREE** + **RANDOM FOREST** =

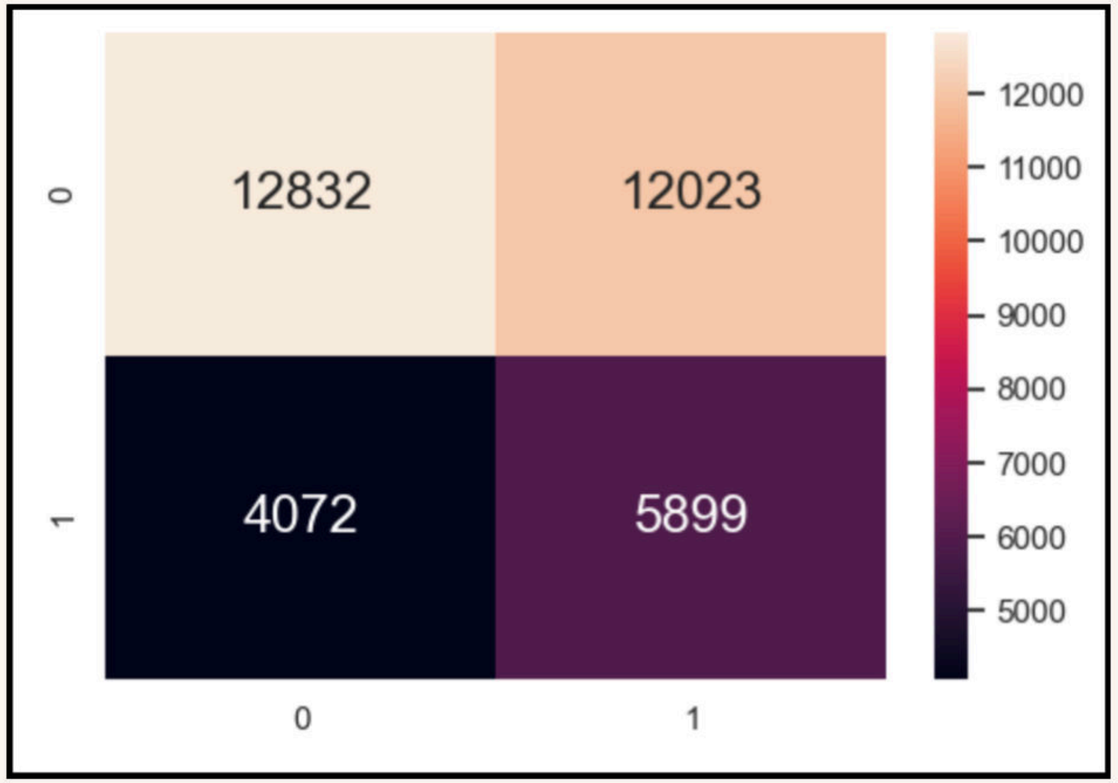


From the Models, the more significant attributes are:

- **CurrentEquipmentDays**
- **MonthsInService**
- **MonthlyMinutes**

Result Evaluation

Logistic Regression

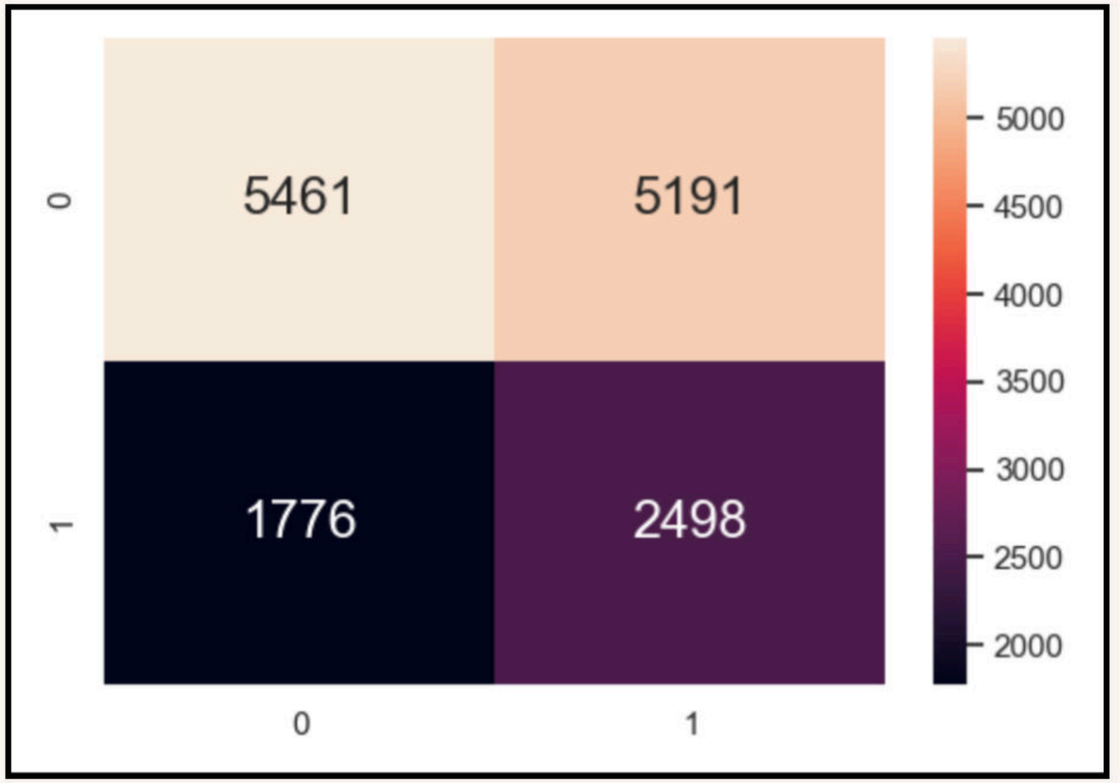


Train Dataset

Classification Accuracy: 71.2%

FPR: 48.4%

FNR: 40.8%



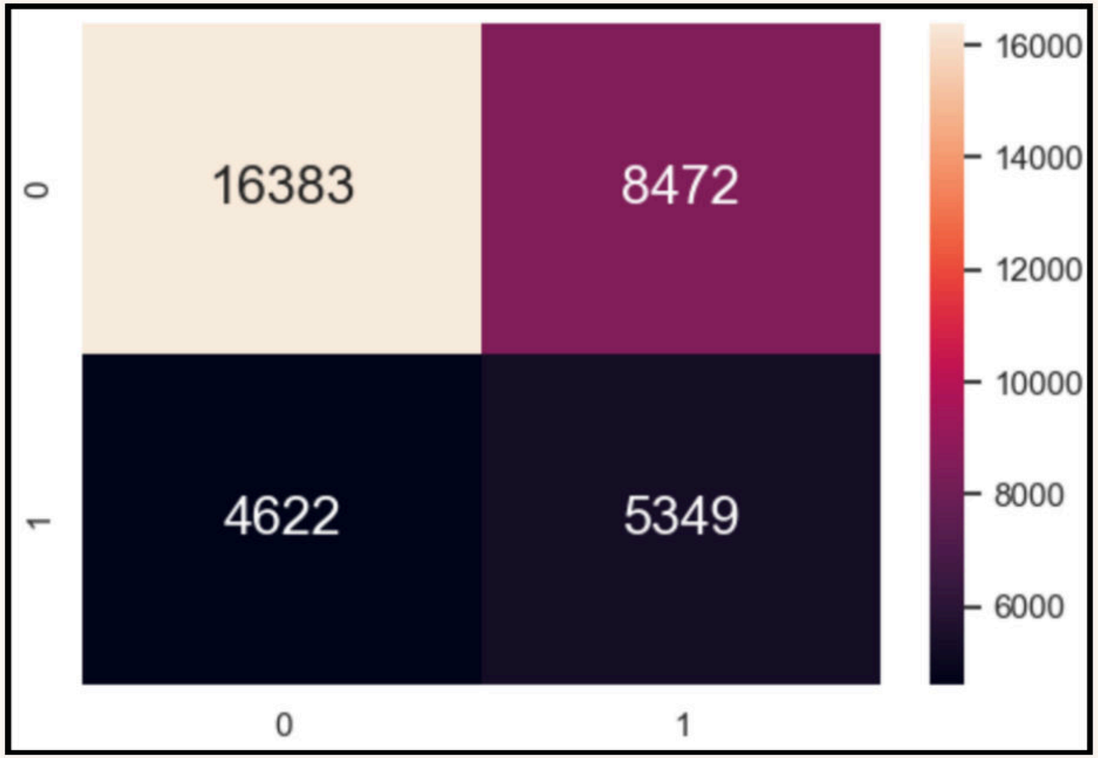
Test Dataset

Classification Accuracy : 71.3%

FPR: 48.7%

FNR: 41.6%

Decision Tree

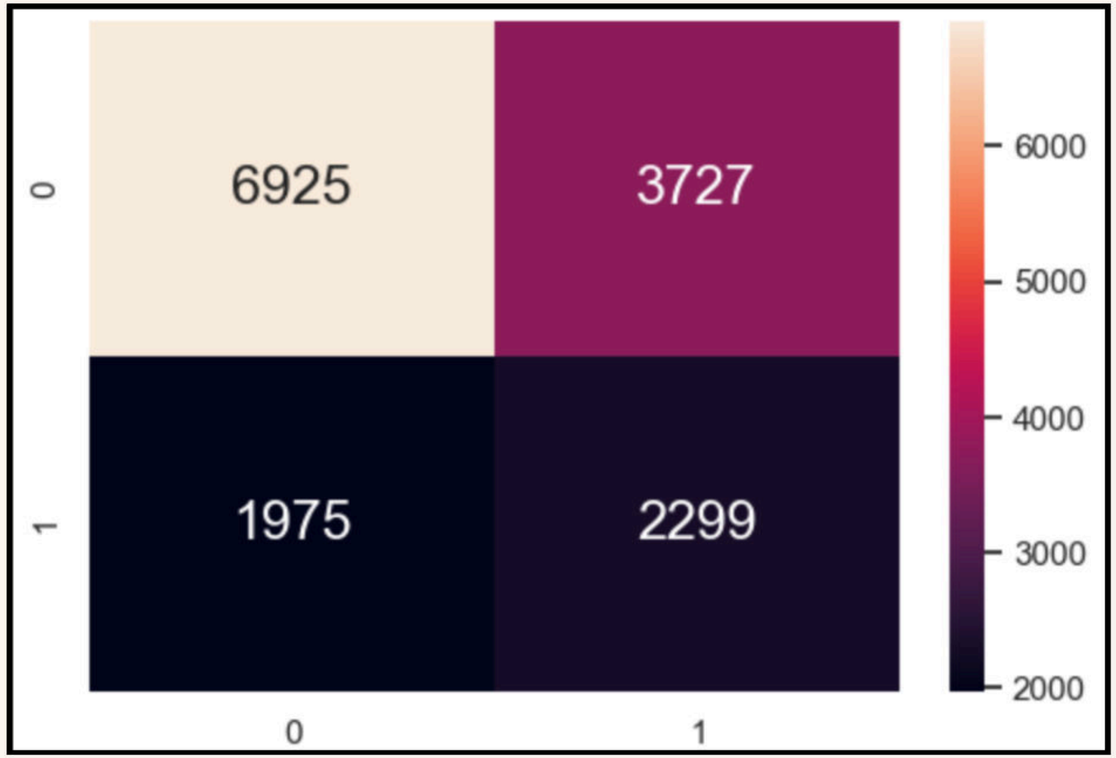


Train Dataset

Classification Accuracy: 71.9%

FPR: 34.1%

FNR: 46.4%



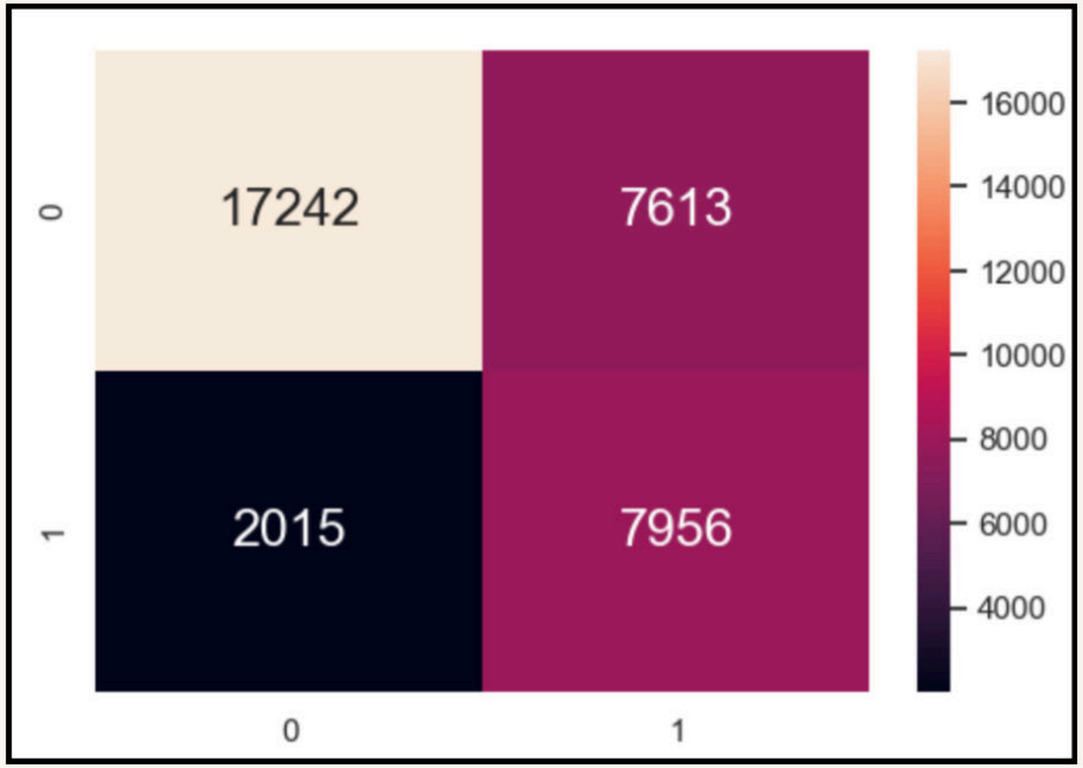
Test Dataset

Classification Accuracy : 71.7%

FPR: 35.0%

FNR: 46.2%

Random Forest

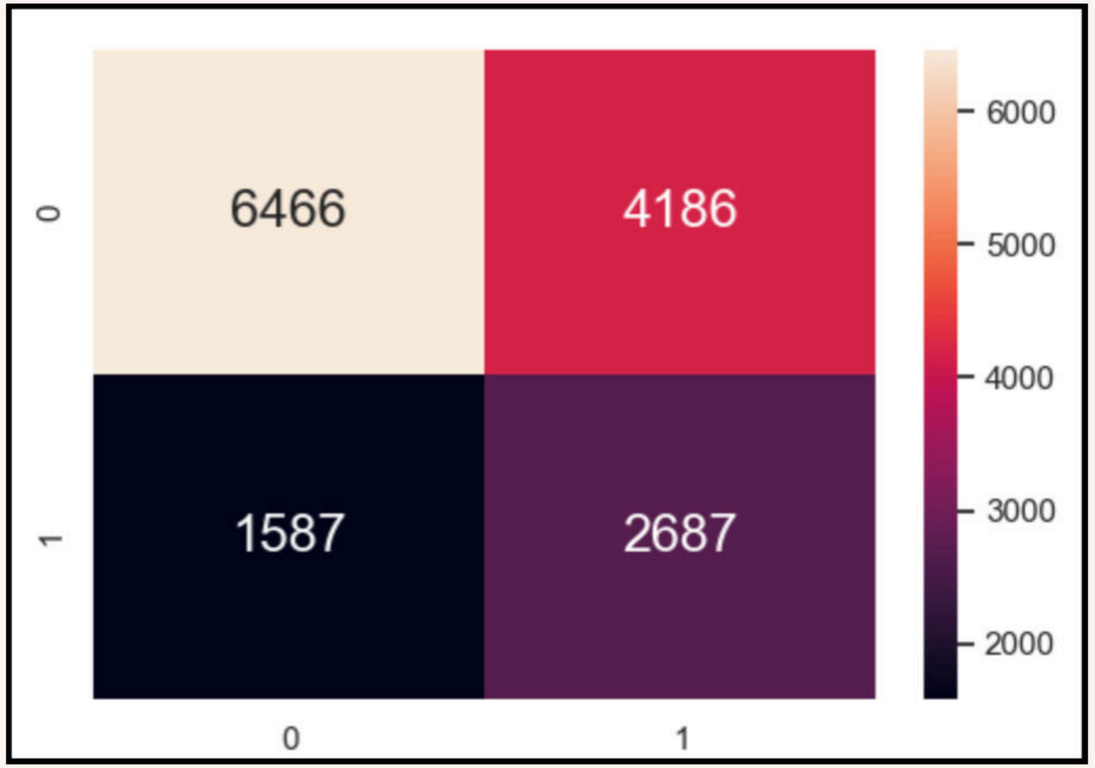


Train Dataset

Classification Accuracy: 74.3%

FPR: 33.3%

FNR: 25.3%

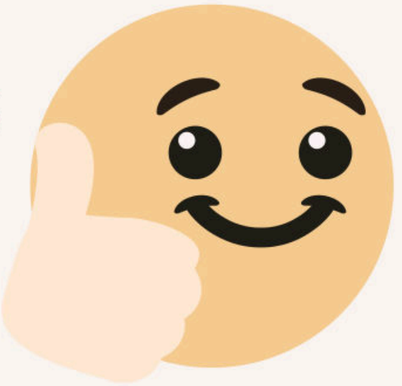


Test Dataset

Classification Accuracy: 72.1%

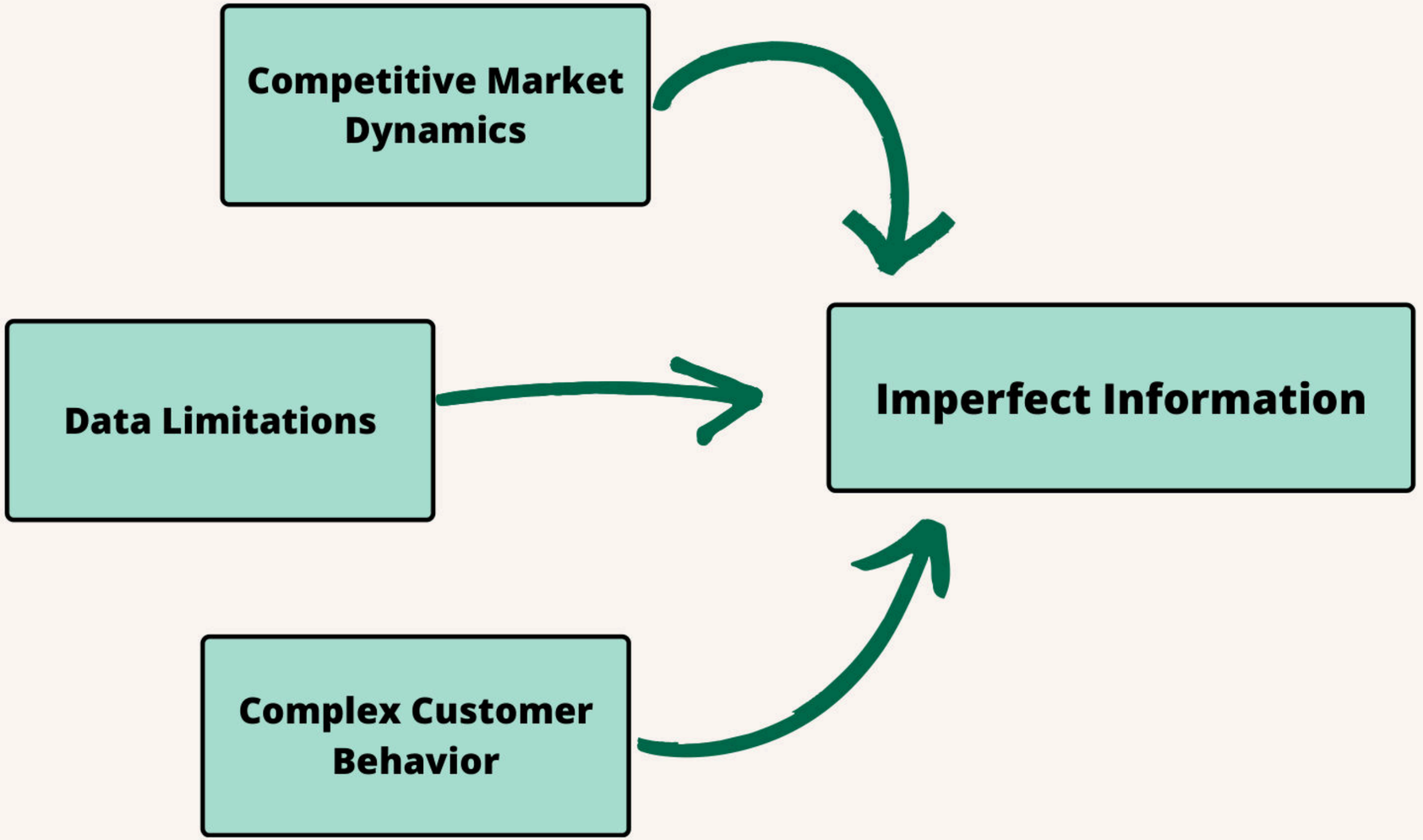
FPR: 39.2%

FNR: 37.1%



Reflection

Possible Reasons



Lesson Learnt

NEW

Logistic Regression & **Random Forest**

Feature Importance



Presented by Group 5

Thank you very much!



References:

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